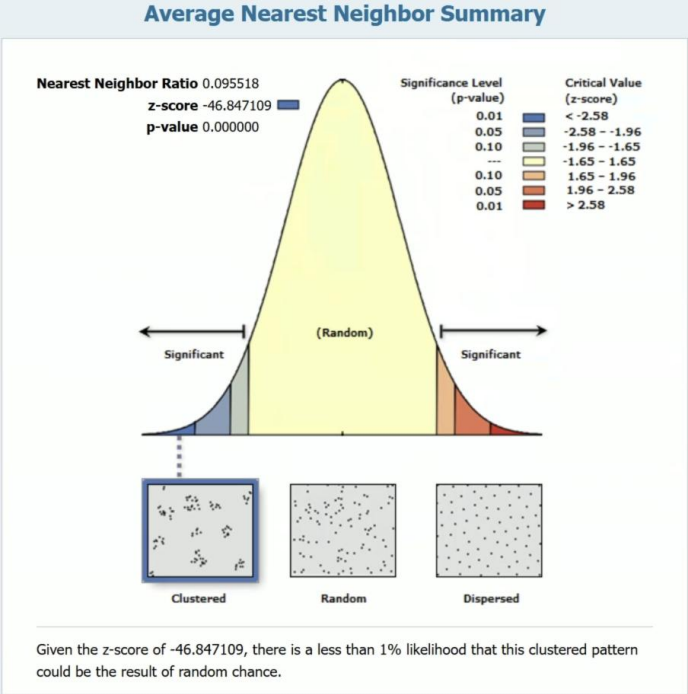
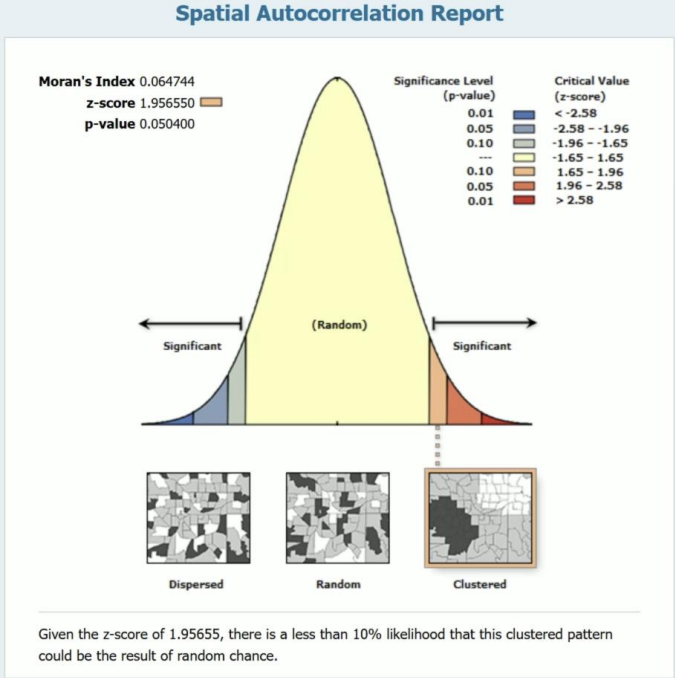
Seattle's West Precinct Burglary Clustering Tests Report

The mean center of the 911 call burglaries in the west precinct of Seattle is southeast of the geographical center of the west precinct. There is a noticeable difference in dispersion between the standard distance circle and the standard deviation ellipse. A big part of this difference is because of the geography of the west precinct. The standard deviation ellipse follows the shape of the land in the west precinct better than the standard deviation circle does. The mean center is in a narrow area of the west precinct peninsula. The standard deviation ellipse minimizes the amount of water within it compared to the standard distance circle. The standard deviation ellipse more closely follows the shape of the land of the west precinct.

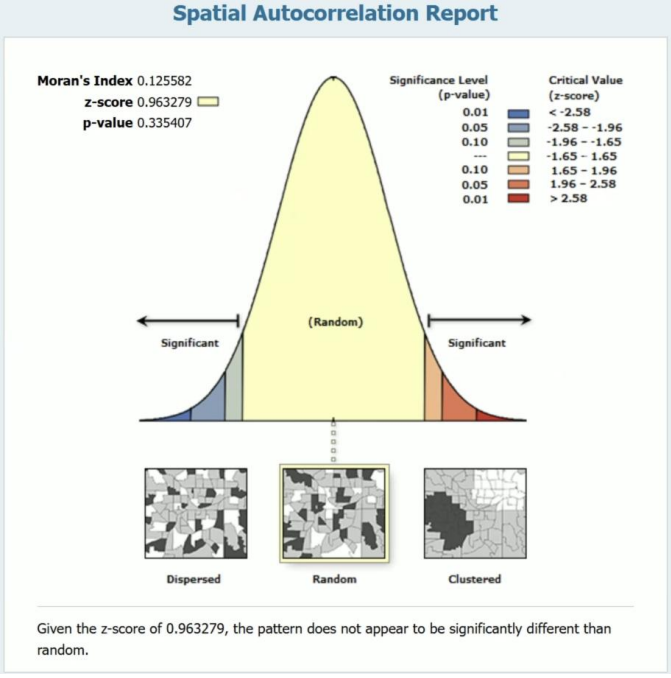
For global spatial statistics, first I ran the average nearest neighbor test. I found that given the z-score of -46.847109, there is a less than 1% likelihood that this clustered pattern could be the result of random chance.



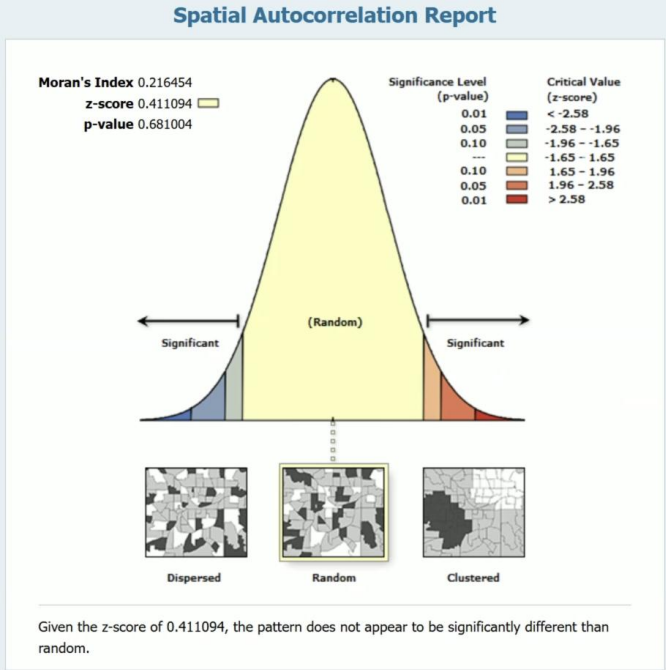
Next I ran a spatial autocorrelation (Global Moran's I) test using inverse distance. I found that given the z-score of 1.95655, there is a less than 10% likelihood that this clustered pattern could be the result of random chance.



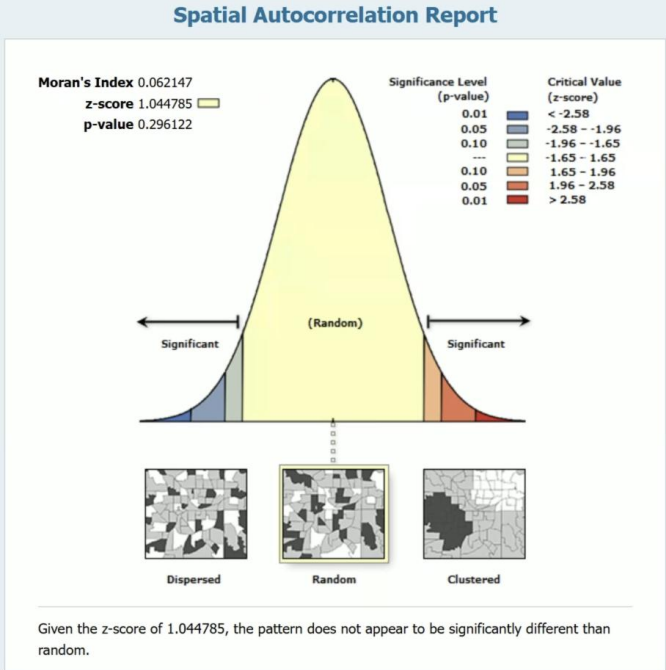
Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 1000 feet. I found that given the z-score of 0.963279, the pattern does not appear to be significantly different than random.



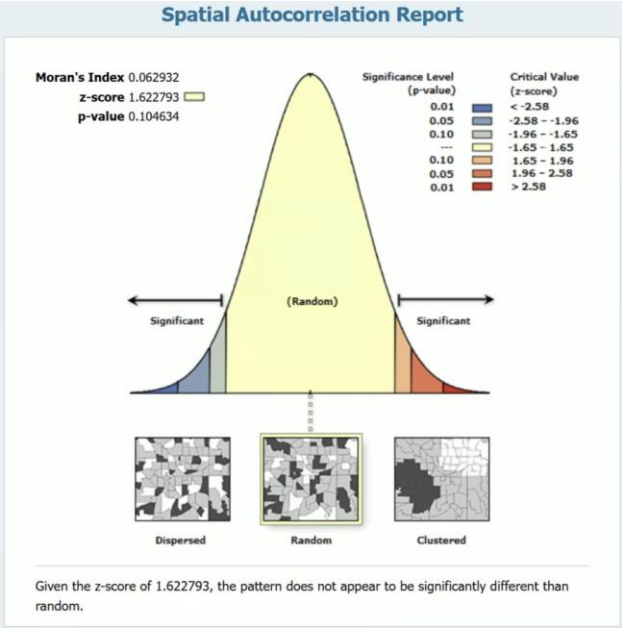
Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 500 feet. I wanted to find out if there might be a clustered pattern in a smaller radius. I found that given the z-score of 0.411094, the pattern does not appear to be significantly different than random.



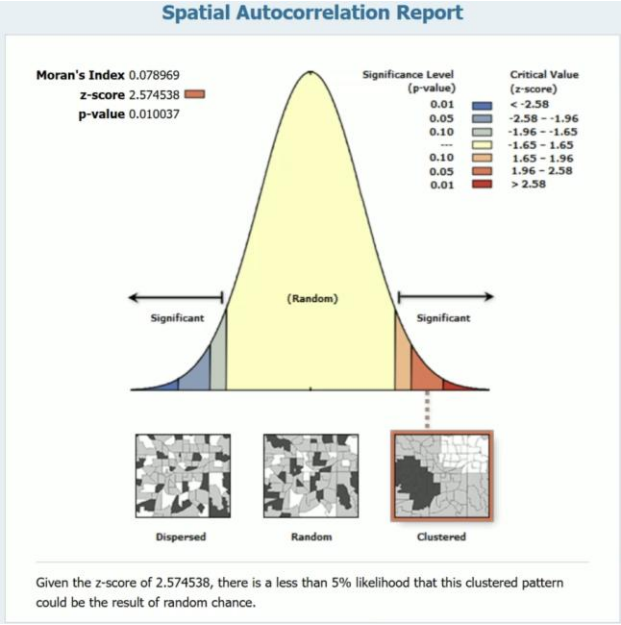
Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 2000 feet. I wanted to find out if there might be a clustered pattern in a larger radius than 1000 feet. I found that given the z-score of 1.044785, the pattern does not appear to be significantly different than random.



Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 3000 feet. I wanted to find out if there might be a clustered pattern in a larger radius than 2000 feet. I found that given the z-score of 1.622793, the pattern does not appear to be significantly different than random.

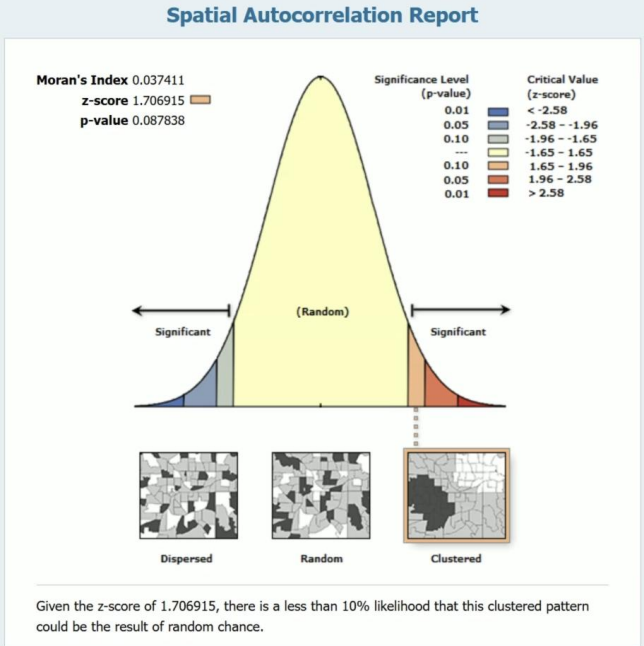


Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 4000 feet. I wanted to find out if there might be a clustered pattern in a larger radius than 3000 feet. I found that given the z-score of 2.574538, there is a less than 5% likelihood that this clustered pattern could be the result of random chance.



Next I ran a spatial autocorrelation (Global Moran's I) test using a fixed distance band of 5000 feet. I wanted to find out if there might be a clustered pattern in a larger radius than 4000 feet. I found that given the z-score of 2.706915, there is a less than 10% likelihood that this clustered pattern could be the result of random chance. This is the same

likelihood that I got when I ran the spatial autocorrelation (Global Moran's I) test using inverse distance, so I stopped here.



At this point, I think that it is a good idea to continue on with the hot spot analysis. The average nearest neighbor test found a less than 1% likelihood that this clustered pattern could be the result of random chance. Both the spatial autocorrelation (Global Moran's I) test using inverse distance and the spatial autocorrelation (Global Moran's I) test using a fixed distance band of 5000 feet found a less than 10% likelihood that this clustered pattern could be the result of random chance. The spatial autocorrelation (Global Moran's I) test using a fixed distance band of 4000 feet found a less than 5% likelihood that this clustered pattern could be the result of random chance. I believe that these global spatial statistics show enough statistical evidence of clustering that it is a good idea to continue on with the hot spot analysis.

After running the global spatial statics, I ran local spatial statistics. The Local Moran's I (LISA) identified hot and cold spots. The Getis-Ord Gi* highlighted statistically significant hot spots, with the hot spots varying from 90% to 99% confidence. The Getis-Ord Gi* results are included on the map.

After running the local spatial statistics, I still believe that it is a good idea to continue on with the hot spot analysis, especially since the Getis-Ord Gi* test showed 99% confidence for some hot spots.